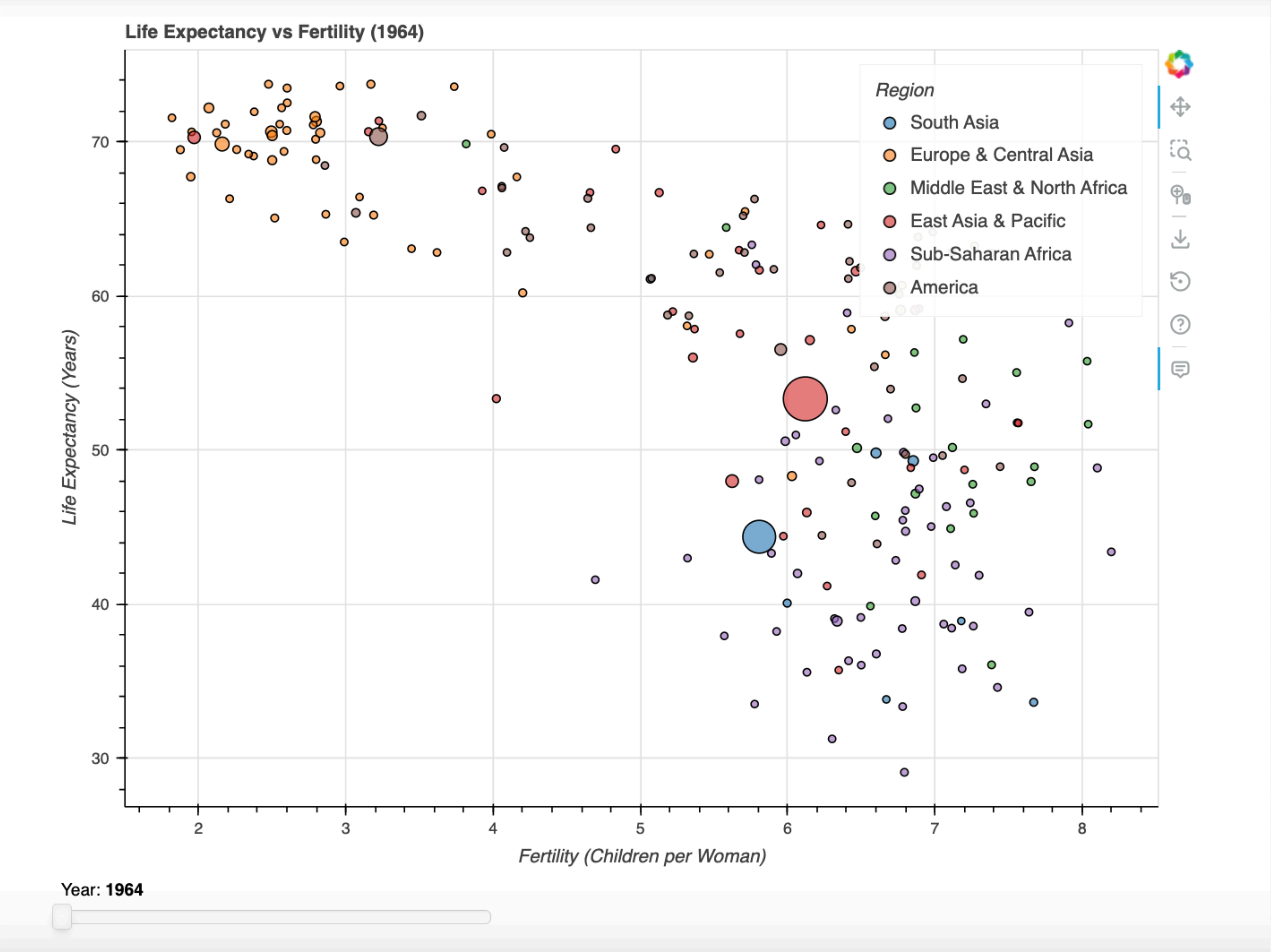
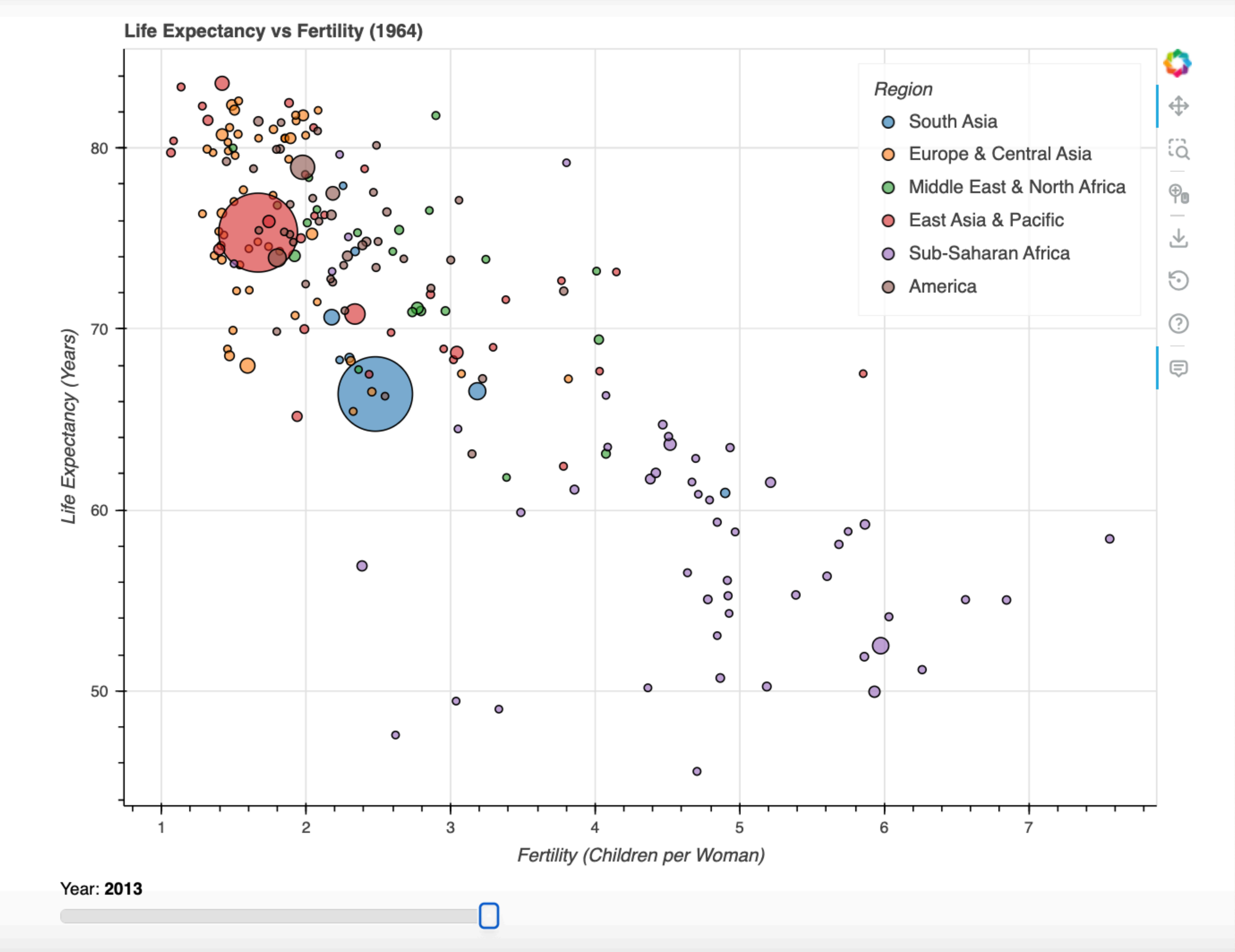




Life Expectancy and Fertility Visualization

Interactive Data Visualization using Python and Bokeh

Rongman Sun

Research Question

Main Question

How did life expectancy and fertility change between 1964 and 2013?

- Which regions developed the fastest?
- Did differences between countries become smaller?

Goal

interactive visualization:

- compare regions
- long-term changes
- global development trends.

Dataset & Design Choice

Dataset

- Gapminder.org
- 244 countries
- 1964–2013
- CSV format

Variables

- Country
- Year
- Region
- Population
- Life expectancy
- Fertility

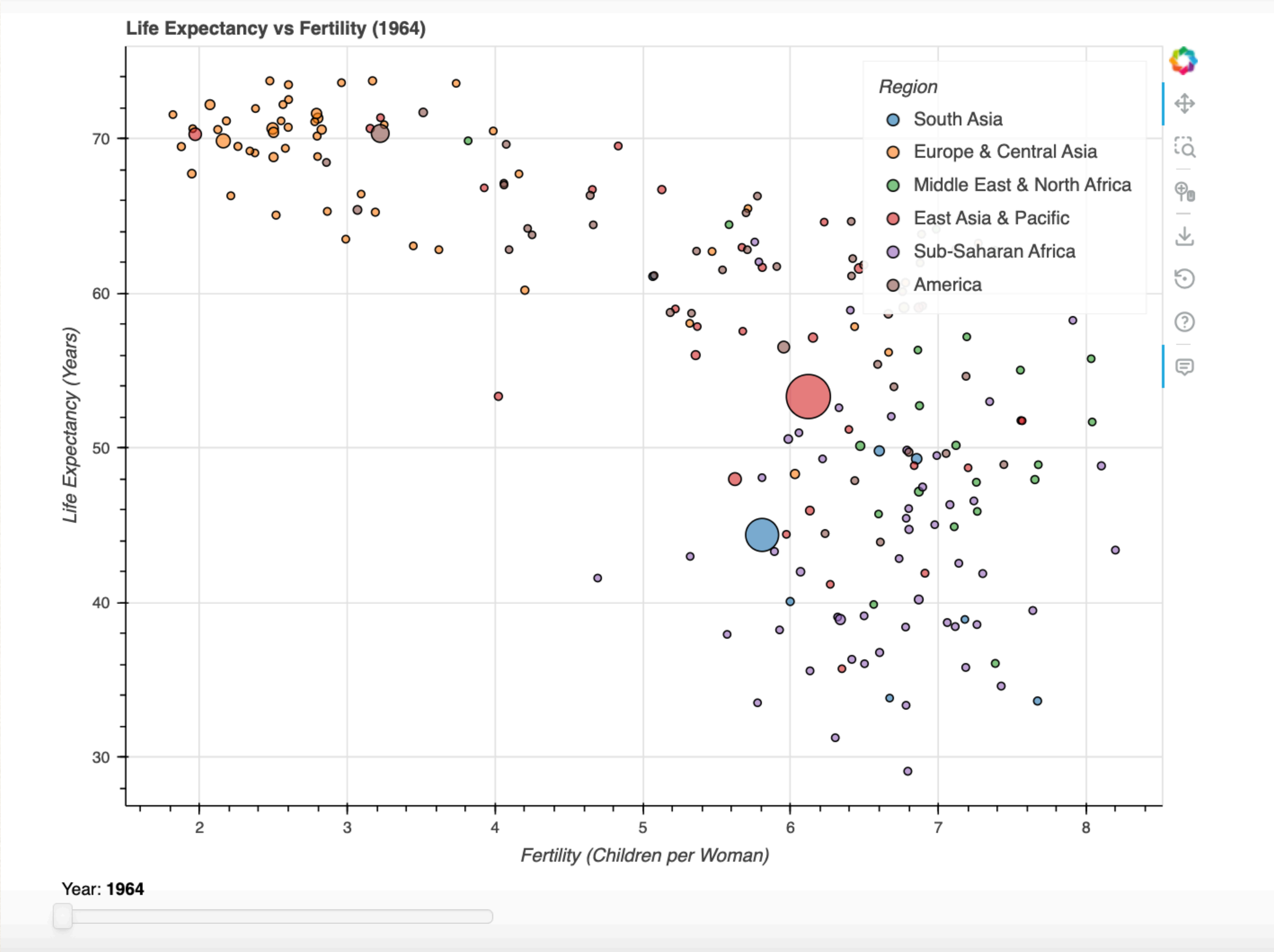
Scatter Plot Design

- X-axis → Fertility
- Y-axis → Life expectancy
- Bubble size → Population
- Color → Region

Purpose

- compare multiple variables
- regional differences
- time changes

```
p.scatter(
    x='x',
    y='y',
    size='pop_size',
    fill_color=palette[i]
)
```



Interactive Features

Bokeh Features

- Year slider
- Hover tooltips
- Clickable legend
- Dynamic updates

Hover Information

- Year
- Country
- Population
- Fertility
- Life expectancy

Interaction Slider + JavaScript callback.

```
# HoverTool
hover = HoverTool(tooltips=[
    ("Country", "@country"),
    ("Year", "@year"),
    ("Life Expectancy", "@y{0.0}"),
    ("Fertility", "@x{0.00}"),
    ("Population", "@population{0,0}"),
])
p.add_tools(hover)

p.legend.title = "Region"
p.legend.location = "top_right"
p.legend.click_policy = "hide"

slider = Slider(start=years[0], end=years[-1], value=initial_year, step=1, title="Year")

# connects the slider with JavaScript callback,chart updates automatically when year changes.
slider.js_on_change('value', callback)
```

Workflow

Python → HTML → JavaScript → BokehJS

Data process

① Python/Pandas (CSV, scale pop, split region)

② Bokeh visualization (plot, slider, legend, hover)

③ User selects a year (slider Interaction)

④ JavaScript callback (filter data source)

⑤ BokehJS redraw (update chart)

Main Learning

- Python + JavaScript + Bokeh
- interactive systems

JavaScript

```javascript
if(data['Year'][j] === year &&
   data['Region'][j] === region)
```

```python
slider.js_on_change('value', callback)
```

Main Findings

Global Trends (chart shift)

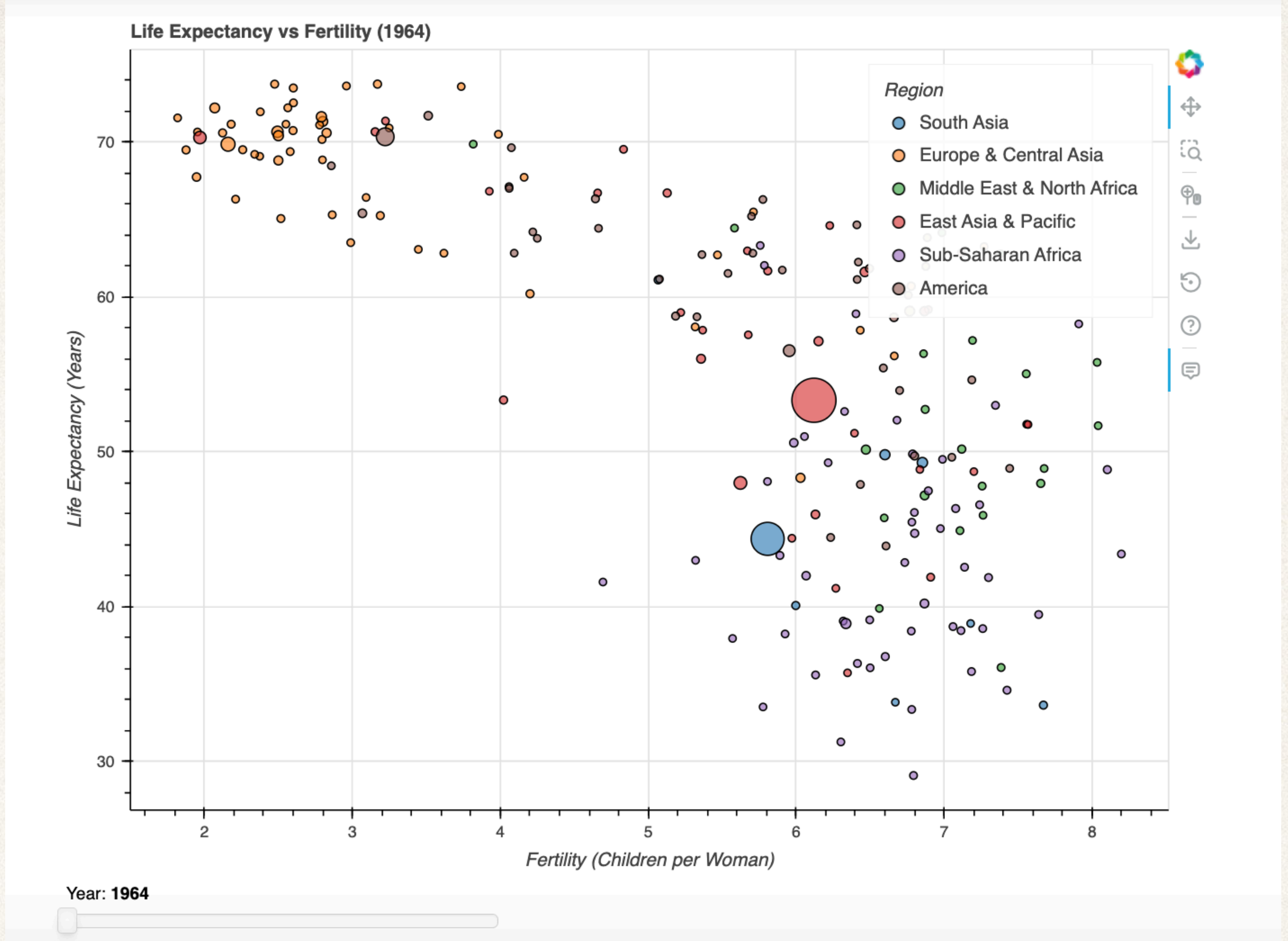
- Fertility ⬇️ Life expectancy ⬆️

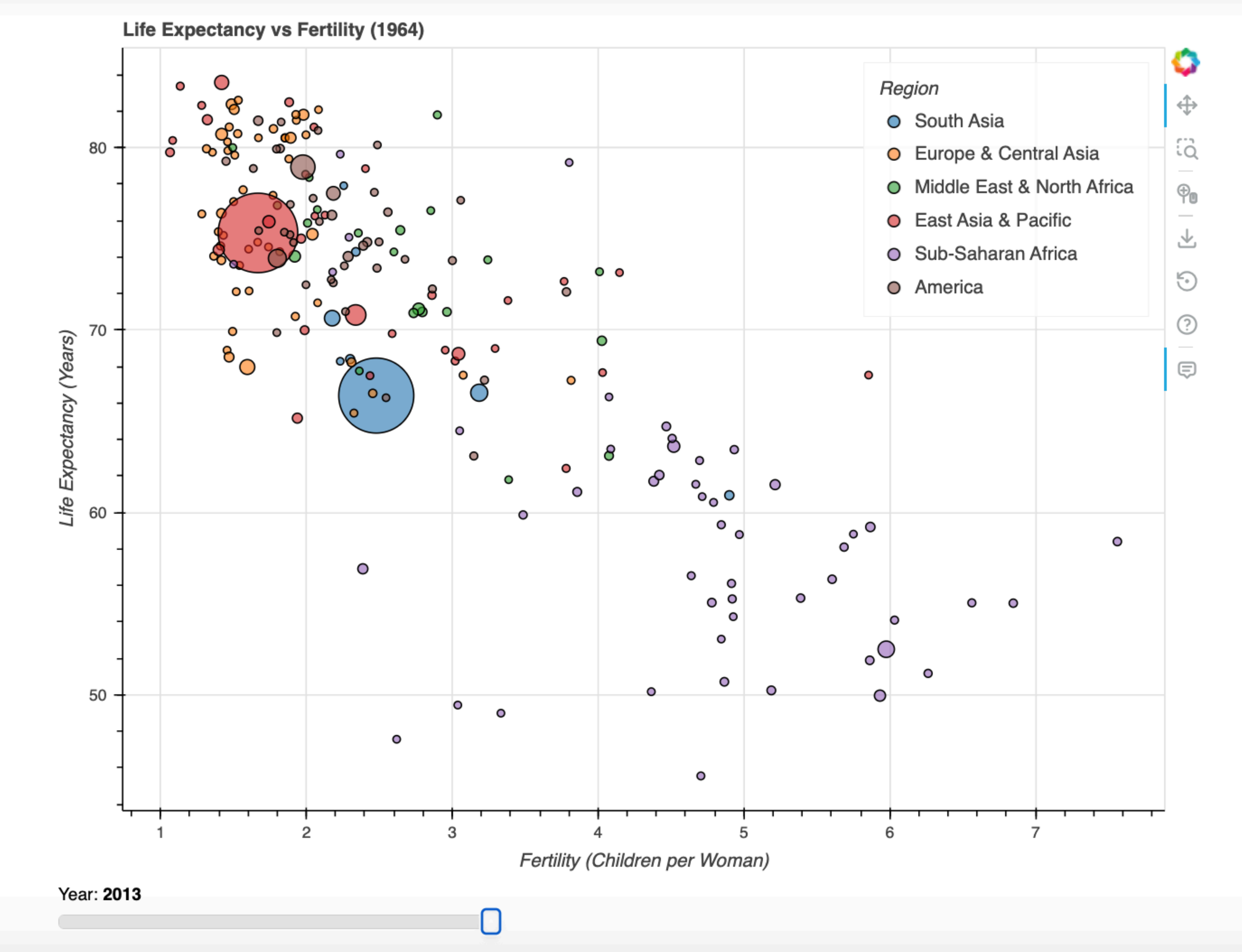
Regional Development (Life expectancy)

- Europe&North America developed earlier
- East Asia rapidly
- Sub-Saharan Africa slowly

Country Differences (points concentrated)

- Countries became similar





Limitations & Future Improvement

Limitations

- Overlapping countries
- Simplified population scaling
- Missing data
- Browser performance limits

Future Improvements

- Country search function
- Animation playback
- Map visualization

Challenges & Learning

Challenges

- First time using Bokeh
- JavaScript callbacks
- Multiple data sources

What I Learned

- Interactive visualization
- Data transformation (read + scale + split)
- Bokeh (Python + Javascript)
- GitHub (code + project)

design choices simplify data, interactive visualization better trends

Thanks